Status	Finished
Started	Tuesday, 20 October 2026, 7:27 PM
Completed	Tuesday, 20 October 2026, 7:31 PM
Duration	4 mins 32 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Quick Sort
- b. Linear Search
- c. Bubble Sort
- d. Merge Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It builds a sorted array one element at a time
- b. It repeatedly finds the minimum element and moves it to the sorted part of the list
- c. It divides the list into two halves, sorts each half, and then merges them
- d. It compares adjacent elements and swaps them if necessary

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Bubble Sort
- b. Quick Sort
- c. Merge Sort
- d. Heap Sort

The correct answer is: Bubble Sort

Question **4**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Dividing the list into halves
- b. Bubbling up the smallest element
- c. Sorting the entire list in one pass
- d. Bubbling up the largest element to its correct position

The correct answer is: Bubbling up the largest element to its correct position

Question **5**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are split again into smaller sublists
- b. They are combined to form a single sorted list
- c. They are compared element by element
- d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **6**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. order()
- b. arrange()
- c. sort()
- d. sorted()

The correct answer is: sort()

Question **7**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. order()
- b. arrange()
- c. sort()
- d. sorted()

The correct answer is: sorted()

Question **8**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Binary Search
- b. Merge Sort
- c. Bubble Sort
- d. Linear Search

The correct answer is: Bubble Sort

Question **9**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Sorting data in a non-sequential manner
- b. Being the most efficient sorting algorithm
- c. Bubbling up the largest element to its correct position with each pass
- d. Using the divide-and-conquer approach

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **10**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Quick sort
b. Merge sort
c. Selection sort
d. Heap sort

The correct answer is:
Merge sort

Question **11**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Combining sorted sublists
b. Sorting the entire list sequentially
c. Comparing each element with the others
d. Dividing the input into smaller subproblems

The correct answer is: Dividing the input into smaller subproblems

Question **12**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting helps in finding duplicates quickly
b. Sorting decreases the efficiency of selection operations
c. Sorting is rarely used in programming
d. Sorting makes it harder to search for items

The correct answer is: Sorting helps in finding duplicates quickly

Question **13**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It has no effect on the analysis process
b. It makes the analysis slower
c. It allows for quicker identification of duplicates
d. It complicates the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **14**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Selection
b. All of the above
c. Duplicates
d. Searching

The correct answer is: All of the above

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Bubble Sort
b. Linear Search
c. Binary Search
d. Merge Sort

The correct answer is: Merge Sort